Signals and Systems — Reference

the big picture

Everything in this course is one of two things:

- a **signal**: a quantity that changes over time, like a voltage or a sound;
- a **system**: a machine that takes a signal in and puts a signal out.

The topics build on each other in this order:

1. **How big is a signal?** Energy and power (4–5).
2. **Building blocks.** The button δ, the switch u, and the ramp r (10).
3. **Machines.** Systems, block diagrams, difference equations (11).
4. **Describing machines.** Memory, causality, stability, linearity, time invariance (12).

The link from 1 to 4: BIBO stability asks if a *bounded* input gives a *bounded* output. That is a question about signal size.

plain-English glossary

Symbol	Say it as
$\delta[n]$, $\delta(t)$	**the button**: pressed once at time 0, then released
$u[n]$, $u(t)$	**the switch**: flipped on at time 0 and left on
$r[n]$, $r(t)$	**the ramp**: the running total of the switch, $0, 1, 2, 3, \ldots$
t	the time variable; it sweeps through every time, and is gone after you integrate
T	the half-width of a window $[-T, T]$; it is the thing that goes to ∞
t_0, n_0	"t-naught": one fixed time, such as 3
N	DT window half-width ($2N+1$ samples), or a DT period
ε	"epsilon": a tiny number that we shrink toward 0
$\int \ldots dt$	add up area as t sweeps (CT)
$\sum$	add up separate terms (DT); the DT twin of $\int$
dt	an infinitely thin slice of time. Not Δt.
B	a ceiling: the input never goes above B in size

notation: CT and DT side by side

Item	Continuous time (CT)	Discrete time (DT)
signal	$x(t)$, round brackets	$x[n]$, square brackets
time variable	t, any real number	n, integers only
period	T (seconds)	N (samples, an integer)
frequency	ω (rad/s), f (Hz)	$\hat{\omega}$ (rad/sample)
energy, power	integral $\int$	sum $\sum$
rate of change	derivative $\frac{d}{dt}$	first difference $x[n]-x[n-1]$
running total	integral $\int_{-\infty}^{t}$	running sum $\sum_{k=-\infty}^{n}$
button	$\delta(t)$, area 1	$\delta[n]$, height 1

sinusoid quantities

Symbol	Name	Unit	Link
A	amplitude	same as x	peak value
ω	angular frequency	rad/s	$\omega = 2\pi f$
f	frequency	Hz = cycles/s	$f = \frac{1}{T}$
T	period	s	$T = \frac{2\pi}{\omega}$
ϕ	phase	rad	$\phi = -\omega t_0$
t_0	time shift	s	$t_0 = -\frac{\phi}{\omega}$
$\hat{\omega}$	DT frequency	rad/sample	$\hat{\omega} = \omega T_s$
T_s, f_s	sampling period, rate	s, Hz	$f_s = \frac{1}{T_s}$

trig and complex number facts

θ	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π
$\cos\theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0	-1
$\sin\theta$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1	0

$\sin 0 = 0$ and $\cos 0 = 1$. Do not swap them.

sine as a cosine	$\sin\theta = \cos\left(\theta - \frac{\pi}{2}\right)$
negative amplitude	$-A\cos\theta = A\cos(\theta+\pi)$
square of a cosine	$\cos^2\theta = \frac{1+\cos(2\theta)}{2}$
Euler's formula	$e^{j\theta} = \cos\theta + j\sin\theta$
cosine from exponentials	$\cos\theta = \frac{e^{j\theta}+e^{-j\theta}}{2}$
sine from exponentials	$\sin\theta = \frac{e^{j\theta}-e^{-j\theta}}{2j}$
size of $a+jb$	$\sqrt{a^2+b^2}$
angle of $a+jb$	$\arctan\frac{b}{a}$ (check the quadrant)
polar form	$C = \lvert C\rvert e^{j\phi}$
size of $e^{j\theta}$	1
whole turns	$e^{j2\pi k} = 1$ for each integer k

sums, integrals, limits

Item	Result
infinite geometric sum, $\lvert r\rvert < 1$	$\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}$
finite geometric sum, $r \neq 1$	$\sum_{n=0}^{N-1} r^n = \frac{1-r^N}{1-r}$
count of terms from $-N$ to N	$2N+1$
$\int t^n\,dt$	$\frac{t^{n+1}}{n+1}$
$\int e^{at}\,dt$	$\frac{e^{at}}{a}$
$\int_0^{\infty} e^{-at}\,dt, \quad a>0$	$\frac{1}{a}$
definite integral	$F(\text{top}) - F(\text{bottom})$
fixed number over a growing T	$\lim_{T\to\infty}\frac{c}{T} = 0$
a growing T alone	$\lim_{T\to\infty} cT = \infty \quad (c>0)$
T cancels	$\frac{cT}{2T} = \frac{c}{2}$, a constant

formula sheet

1. Signals and sampling

sampling	$x[n] = x(nT_s)$
sampling rate	$f_s = \frac{1}{T_s}$

2–3. Sinusoids

CT sinusoid	$x(t) = A\cos(\omega t + \phi)$
angular frequency	$\omega = 2\pi f$
period	$T = \frac{1}{f} = \frac{2\pi}{\omega}$
phase → time shift	$t_0 = -\frac{\phi}{\omega}$
time shift → phase	$\phi = -\omega t_0$
DT sinusoid	$x[n] = A\cos(\hat{\omega} n + \phi)$
CT → DT frequency	$\hat{\omega} = \omega T_s = \frac{2\pi f}{f_s}$
DT → CT frequency	$f = \frac{\hat{\omega} f_s}{2\pi}$
aliases	$\hat{\omega} + 2\pi k, \quad f + k f_s$

4–5. Energy and power

CT energy (total)	$E_\infty = \int_{-\infty}^{\infty} \lvert x(t)\rvert^2\,dt$
DT energy (total)	$E_\infty = \sum_{n=-\infty}^{\infty} \lvert x[n]\rvert^2$
CT power (average)	$P_\infty = \lim_{T\to\infty} \frac{1}{2T}\int_{-T}^{T} \lvert x(t)\rvert^2\,dt$
DT power (average)	$P_\infty = \lim_{N\to\infty} \frac{1}{2N+1}\sum_{n=-N}^{N} \lvert x[n]\rvert^2$
periodic CT power	$P = \frac{1}{T}\int_{\text{one period}} \lvert x(t)\rvert^2\,dt$
periodic DT power	$P = \frac{1}{N}\sum_{\text{one period}} \lvert x[n]\rvert^2$
sinusoid power	$P = \frac{A^2}{2}$
constant c power	$P = c^2$
flat piece, level A, length D	$A^2 D$
ramp $0 \to A$, length D	$\frac{A^2 D}{3}$
line $A \to B$, length D	$\frac{D}{3}\left(A^2 + AB + B^2\right)$
DT stem of height h	h^2

6. Transformations

shift	$x(t-t_0)$: right (delay) if $t_0 > 0$
reversal	$x(-t)$: mirror about $t = 0$
scaling	$x(at)$: compress if $\lvert a\rvert > 1$
point map for $x(at+b)$	$t_{\text{new}} = \frac{t_{\text{old}} - b}{a}$
undo $y(t) = x(at+b)$	$x(t) = y\left(\frac{t-b}{a}\right)$

7. Periodic signals

CT test	$x(t+T) = x(t)$
DT test	$x[n+N] = x[n]$, N an integer
DT sinusoid period	$\frac{\hat{\omega}}{2\pi} = \frac{m}{N}$ in lowest terms
sum of two periodic signals	lcm of the two periods

8. Even and odd

even test	$x(-t) = x(t)$
odd test	$x(-t) = -x(t)$
even part	$\mathcal{E}v\{x\} = \frac{x(t) + x(-t)}{2}$
odd part	$\mathcal{O}d\{x\} = \frac{x(t) - x(-t)}{2}$

9. Complex exponentials

CT form	$x(t) = Ce^{at}, \quad a = \sigma + j\omega$
real part	$\lvert C\rvert e^{\sigma t}\cos(\omega t + \phi), \quad \phi = \angle C$
DT form	$x[n] = C\alpha^n$

10. Button, switch, ramp

DT switch → button	$\delta[n] = u[n] - u[n-1]$
DT button → switch	$u[n] = \sum_{k=-\infty}^{n} \delta[k]$
CT switch → button	$\delta(t) = \frac{du(t)}{dt}$
CT button → switch	$u(t) = \int_{-\infty}^{t} \delta(\tau)\,d\tau$
area of the CT button	$\int_{-\infty}^{\infty} \delta(t)\,dt = 1$
sifting (the flash)	$\int_{-\infty}^{\infty} x(t)\,\delta(t-t_0)\,dt = x(t_0)$
DT sifting	$\sum_n x[n]\,\delta[n-n_0] = x[n_0]$
CT pulse from switches	$u(t-t_1) - u(t-t_2)$: on for $t_1 \le t < t_2$
DT pulse from switches	$u[n-a] - u[n-b]$: on for $a \le n \le b-1$
any DT signal as buttons	$x[n] = \sum_k x[k]\,\delta[n-k]$
ramp	$r(t) = t\,u(t), \quad r[n] = n\,u[n]$
ramp and switch, CT	$\frac{dr}{dt} = u(t)$
ramp and switch, DT	$r[n] - r[n-1] = u[n-1]$
graph from ramps	corner: $+(\Delta\text{slope})\,r(t-t_0)$

11–12. DT systems

unit delay D	$x[n] \to x[n-1]$
first difference	$y[n] = x[n] - x[n-1]$
3-point moving average	$y[n] = \frac{1}{3}\left(x[n] + x[n-1] + x[n-2]\right)$
accumulator	$y[n] = y[n-1] + x[n]$
bounded input	$\lvert x[n]\rvert \le B < \infty$
stable if	$\lvert y[n]\rvert \le$ a finite number

system property tests

Property	Ask yourself	Yes / no example
memoryless	Does y now use only x now?	yes x^2 no $x[n-1]$
causal	Does y avoid peeking at the future?	yes $x[n-1]$ no $x[n+1]$
stable	Does a capped input give a capped output?	yes x^2 no $\sum_{k\le n} x[k]$
invertible	Can I undo it and get x back?	yes $2x$ no x^2
time invariant	Does it ignore the clock? Delay in = delay out	yes $x[n-1]$ no $nx[n]$
linear	Scale in → scale out; add in → add out?	yes $nx[n]$ no $2x+3$

which tool? — given → find

Given → find	Do this	
$x(t)$, $T_s \to x[n]$	put $t = nT_s$	(1)
$A\cos(\omega t + \phi) \to f, T, t_0$	read, then divide	(2)
$f, f_s \to \hat{\omega}$	$\hat{\omega} = 2\pi f / f_s$	(3)
dies out or finite length	energy E_∞	(4)
periodic or constant	power P_∞	(4)
a graph → energy	square each piece, add	(5)
$x(t) \to x(at+b)$	map each breakpoint	(6)
$y = x(at+b) \to x$	$x(t) = y((t-b)/a)$	(6)
two graphs → $y = Ax(at+b)$	match two points, solve	(6)
signal → period	$2\pi/\omega$, or reduce $\hat{\omega}/2\pi$	(7)
signal → even, odd, neither	compare $x(-t)$ with $x(t)$	(8)
Ce^{at} → real signal	Euler, then the real part	(9)
integral with $\delta(t-t_0)$	the flash: three questions	(10)
a pulse graph → formula	a switch on, a switch off	(10)
DT stems → buttons	one $x[k]\,\delta[n-k]$ per stem	(10)
graph with corners → formula	a ramp per corner	(10)
blocks → one equation	substitute block by block	(11)
equation ↔ block diagram	one D per shift	(11)
system → properties	ask each question	(12)
system → stable?	cap the input with B	(12)

The number in parentheses is the concept section.

Signals and Systems — Concepts

Puzzle pieces. In every worked example, a colored piece read off the formula or graph goes in the slot [of that color] in the template. Order: **pieces → template → place → check**. Sinusoid colors stay the same everywhere: amplitude, frequency, phase or shift.

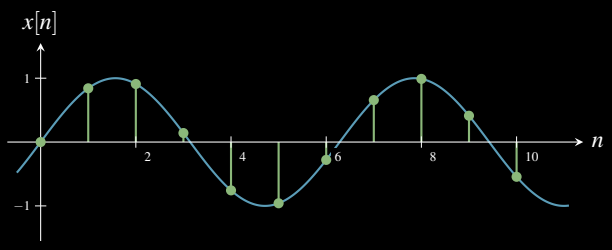
1. signals and sampling

THE IDEA
A **signal** is anything that changes over time and carries information. **Sampling** is taking snapshots: one photo every T_s seconds, like a strobe light.

- **Continuous time** (CT): $x(t)$, round brackets, any real t. The movie.
- **Discrete time** (DT): $x[n]$, square brackets, n an integer only. The stack of photos.

Trap: $x[2.5]$ does not exist. Between samples a DT signal is not zero; it is undefined.

Movie to photos: photo n is taken at $t = nT_s$, so $x[n] = x(nT_s)$. T_s = **sampling period** (s between photos), $f_s = \frac{1}{T_s}$ = **sampling rate** (photos per s).



The curve is the movie (CT). The stems are the photos (DT).

Worked: sample $x(t) = t^2$ with $T_s = 0.5$ s

1. PIECES
the movie $x(t) = t^2$ the photo spacing $T_s = 0.5$

↓

2. TEMPLATE
$x[n] = [x(\cdot)]$ evaluated at $t = n \cdot [T_s]$

↓

3. PLACE
$x[n] = (n \cdot 0.5)^2 = 0.25n^2$ $x[0..3] = 0,\ 0.25,\ 1,\ 2.25$

↓

4. CHECK
$n = 2$ is $t = 1$ s; $x(1) = 1$ ✓

Worked: (reverse) $x[n] = 0.25n^2$ came from $T_s = 0.5$ s; find $x(t)$

1. TEMPLATE
$x(t) = x[n]$ with $n = \dfrac{t}{[T_s]}$

↓

2. PLACE
$n = \dfrac{t}{0.5} = 2t \quad \Rightarrow \quad x(t) = 0.25(2t)^2 = t^2$

2. sinusoids

THE IDEA
A wheel spins at a steady speed with a dot on its rim. The dot's height over time is a sinusoid. Wheel size: amplitude. Spin rate: frequency. Where the dot starts: phase.

$$x(t) = [A]\cos\left([\omega]t + [\phi]\right)$$

- A: **amplitude**, the peak; swings between $-A$ and A.
- ω: **angular frequency**, rad/s. $f = \frac{\omega}{2\pi}$ Hz (turns per s), $T = \frac{1}{f} = \frac{2\pi}{\omega}$ s per turn.
- ϕ: **phase**, rad. A head start or late start. As a **time shift**: $t_0 = -\frac{\phi}{\omega}$, $A\cos\left(\omega(t - t_0)\right)$.

Negative $\phi \Rightarrow t_0 > 0$: **delayed** (moves right).

Trap: Get the exact form $A\cos(\omega t + \phi)$ before reading pieces. Sine: $\sin\theta = \cos(\theta - \frac{\pi}{2})$. Negative A: $-A\cos\theta = A\cos(\theta + \pi)$.

Which way? — between the sinusoid numbers

Given → find	Do this
$\omega \to f$	divide by 2π: $f = \dfrac{\omega}{2\pi}$
$f \to \omega$	multiply by 2π: $\omega = 2\pi f$
$f \to T$	reciprocal: $T = \dfrac{1}{f}$
$T \to \omega$	$\omega = \dfrac{2\pi}{T}$
$\phi \to t_0$	$t_0 = -\dfrac{\phi}{\omega}$
$t_0 \to \phi$	$\phi = -\omega t_0$

Worked: read the parts of $x(t) = 3\cos\left(4\pi t - \frac{\pi}{3}\right)$

1. PIECES (MATCH THE FORM)
$A = 3$ $\omega = 4\pi$ $\phi = -\frac{\pi}{3}$

↓

2. TEMPLATES
$f = \dfrac{[\omega]}{2\pi}$ $T = \dfrac{2\pi}{[\omega]}$ $t_0 = -\dfrac{[\phi]}{[\omega]}$

↓

3. PLACE
$f = \dfrac{4\pi}{2\pi} = 2$ Hz $T = 0.5$ s $t_0 = -\dfrac{-\pi/3}{4\pi} = \frac{1}{12}$ s *(delayed)*

↓

4. CHECK
$3\cos\left(4\pi(t - \frac{1}{12})\right)$: $4\pi \cdot \frac{1}{12} = \frac{\pi}{3}$ ✓

Worked: (sine first) read the parts of $x(t) = 4\sin(10t)$

1. PREP
$\sin\theta = \cos(\theta - \frac{\pi}{2}) \quad \Rightarrow \quad 4\cos\left(10t - \frac{\pi}{2}\right)$

↓

2. PIECES, PLACE
$A = 4$, $\omega = 10$, $\phi = -\frac{\pi}{2}$ $f = \frac{10}{2\pi} = 1.59$ Hz, $T = \frac{2\pi}{10} = 0.628$ s

Worked: (reverse) build $x(t)$: $A = 2$, $T = 0.4$ s, first peak at $t = 0.1$ s

1. PIECES
$A = 2$ period $0.4 \to \omega = \frac{2\pi}{0.4} = 5\pi$ first peak = shift: $t_0 = 0.1 \to \phi = -5\pi(0.1) = -\frac{\pi}{2}$

↓

2. PLACE IN $[A]\cos([\omega]t + [\phi])$
$x(t) = 2\cos\left(5\pi t - \frac{\pi}{2}\right)$

↓

3. CHECK
angle at $t = 0.1$: $\frac{\pi}{2} - \frac{\pi}{2} = 0 \Rightarrow 2\cos 0 = 2$, the peak ✓

3. DT sinusoids and DT frequency

THE IDEA
Photograph the spinning wheel with a strobe. In the photos you only see *how far the wheel turned between photos*. That angle per photo is the **DT frequency** $\hat{\omega}$, in rad/sample.

$$x[n] = A\cos(\hat{\omega}n + \phi), \qquad \hat{\omega} = [\omega]T_s = \frac{2\pi[f]}{[f_s]}$$

Aliasing is the wagon-wheel effect: a turn of $\frac{\pi}{4}$ per photo looks identical to $\frac{\pi}{4} + 2\pi$. So $\hat{\omega}$ and $\hat{\omega} + 2\pi$ give the same photos; f and $f + f_s$ give the same samples.

Which way? — CT frequency or DT frequency

Given → find	Do this
$f, f_s \to \hat{\omega}$	$\hat{\omega} = \dfrac{2\pi f}{f_s}$
$\omega, T_s \to \hat{\omega}$	$\hat{\omega} = \omega T_s$
$\hat{\omega}, f_s \to f$	$f = \dfrac{\hat{\omega} f_s}{2\pi}$ (the lowest answer)

Trap: DT to CT has more than one answer. Any alias $f + k f_s$ fits the same photos.

Worked: sample $x(t) = \cos(200\pi t)$ at $f_s = 800$ Hz

1. PIECES
$\omega = 200\pi$ ($f = 100$ Hz) $\quad f_s = 800$

↓

2. TEMPLATE
$\hat{\omega} = \dfrac{\boxed{\omega}}{\boxed{f_s}}$ *(same as ωT_s)*

↓

3. PLACE
$\hat{\omega} = \dfrac{200\pi}{800} = \dfrac{\pi}{4}$ rad/sample $\Rightarrow \quad x[n] = \cos\left(\frac{\pi}{4} n\right)$

Worked: (reverse and alias) $\hat{\omega} = \frac{\pi}{4}$, $f_s = 800$; and does 900 Hz match?

1. TEMPLATE
$f = \dfrac{\boxed{\hat{\omega}} \cdot \boxed{f_s}}{2\pi}$

↓

2. PLACE
$f = \dfrac{(\pi/4) \cdot 800}{2\pi} = 100$ Hz

↓

3. ALIAS
900 Hz: $\hat{\omega} = \frac{2\pi \cdot 900}{800} = \frac{9\pi}{4}$; $\quad -2\pi \to \frac{\pi}{4} \quad \Rightarrow$ same samples *($900 = 100 + f_s$)*

4. energy and power

THE IDEA
Think of $|x|^2$ as **money spent per day**, forever. **Energy** is your total spending; **power** is your average per day. Spend \$100 once: total finite, average 0. Spend \$5 every day: total ∞, average \$5.

Why square? Picture x as a voltage on a $1\,\Omega$ resistor: $P = V^2/R = x^2$. Squaring also makes negatives count: $(-0.5)^2 = 0.25$. Without it, $\cos t$ would average to 0.
Total energy: add $|x|^2$ over all time, no dividing.

$$E_\infty = \int_{-\infty}^{\infty} |x(t)|^2\,dt \qquad E_\infty = \sum_{n=-\infty}^{\infty} |x[n]|^2$$

Average power: total inside the window $[-T, T]$, divide by its length $2T$, grow the window.

$$P_\infty = \lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} |x(t)|^2\,dt \qquad P_\infty = \lim_{N\to\infty} \frac{1}{2N+1} \sum_{n=-N}^{N} |x[n]|^2$$

Periodic shortcut: one period tells the story, $P = \frac{1}{T}\int_{\text{period}} |x|^2\,dt$.

Puzzle template for energy: $E = \int_{\boxed{\text{on at}}}^{\boxed{\text{off at}}} \boxed{|x|^2}\,dt$ (DT: $\sum$). Where $x = 0$ it adds nothing, so the limits shrink to where the signal is on; a switch u tells you the on time.

Classify by shape first

The signal	Result
dies out, or finite length	**energy signal**: E_∞ finite, $P_\infty = 0$
periodic, or constant	**power signal**: P_∞ finite, $E_\infty = \infty$
grows without limit ($x = t$)	**neither**: both ∞

Energy signal: a fixed total E over a growing window, $\frac{E}{2T} \to 0$. Constant: energy grows as fast as the window, so T cancels: $\frac{c^2 \cdot 2T}{2T} = c^2$. A final E_∞ or P_∞ never contains T or N.

Traps from practice

Trap: **Square first**, then add. $E_\infty = 0$ only if $x = 0$ everywhere.
Trap: "16 forever" is the **power**; the energy is ∞.
Trap: t disappears when you integrate; T is what goes to ∞. $\quad \frac{c}{T} \to 0$, not c. Write dt, not Δt.
Trap: **Write the label**: energy signal, power signal, or neither.

Worked: pulse: $x(t) = 2$ for $0 \le t \le 3$, else 0

1. PIECES
on at 0, off at 3, $\quad |x|^2 = 4$

↓

2. PLACE, SOLVE
$E = \int_0^3 4\,dt = 12$ *(check $A^2 D = 4 \cdot 3$)* $\qquad P = \lim_{T\to\infty} \frac{12}{2T} = 0 \Rightarrow$ energy signal

Worked: decaying exponential: $x(t) = 3e^{-2t}u(t)$

1. PIECES
switch $u(t) \Rightarrow$ on at 0; never off: ∞; $\quad |x|^2 = 9e^{-4t}$ *(square: double the exponent)*

↓

2. PLACE
$E = \int_0^\infty 9e^{-4t}\,dt$

↓

3. SOLVE (SHORTCUT $\int_0^\infty e^{-at}dt = \frac{1}{a}$)
$E = 9 \cdot \frac{1}{4} = 2.25 \Rightarrow$ energy signal

Worked: DT geometric: $x[n] = \left(\frac{1}{2}\right)^n u[n]$

1. PIECES
on at 0, off never; $\quad |x|^2 = \left(\frac{1}{2}\right)^{2n} = \left(\frac{1}{4}\right)^n$

↓

2. PLACE
$E = \sum_0^\infty \left(\frac{1}{4}\right)^n$ *(geometric, $r = \frac{1}{4}$)*

↓

3. SOLVE, CHECK
$\frac{1}{1-r} = \frac{4}{3} \qquad 1 + 0.25 + 0.0625 + \ldots \approx 1.33$ ✓

Worked: sinusoid: $x(t) = 4\cos(3t)$

1. PIECES
$A = 4 \qquad |x|^2 = 16\cos^2(3t) = 8 + 8\cos(6t)$ *($\cos^2\theta = \frac{1+\cos 2\theta}{2}$)*

↓

2. AVERAGE EACH TERM
8 averages to 8; $\quad 8\cos(6t)$ averages to 0 $\quad \Rightarrow P = 8$

↓

3. CHECK (SHORTCUT $\frac{\boxed{A}^2}{2}$)
$\frac{4^2}{2} = 8$ ✓ $\quad E = \infty \Rightarrow$ power signal

Worked: constant $x(t) = 3$; and DT $x[n] = (-1)^n$

1. WINDOW TOTAL
$\int_{-T}^{T} 9\,dt = 18T$ *(minus a negative: $9T - (-9T)$)*

↓

2. ENERGY vs POWER
$E = \lim_{T\to\infty} 18T = \infty; \qquad P = \lim_{T\to\infty} \frac{18T}{2T} = 9$ *(T cancels; $= c^2$)* $\Rightarrow$ power signal

↓

3. DT
$|(-1)^n|^2 = 1; \quad 2N+1$ terms $\Rightarrow P = \dfrac{2N+1}{2N+1} = 1$

5. energy from a graph

THE IDEA
Do not set up one big integral. Chop the graph into pieces, find each piece's energy from the shortcut table, and add. In DT there are no widths: each stem's energy is its height squared.

Shortcuts (piece of length D) — these are the templates:

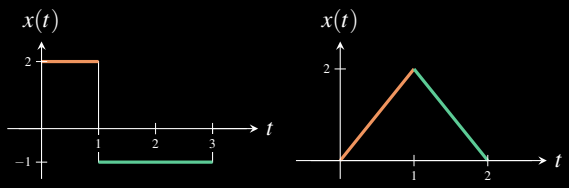
Piece of the graph	Energy of that piece
flat at level A	A^2D
ramp from 0 to A (or A to 0)	$\frac{A^2D}{3}$
line from A to B	$\frac{D}{3}(A^2+AB+B^2)$
DT stem of height h	h^2

Line through (t_1,x_1), slope m: $x = x_1 + m(t-t_1)$. Use a shortcut to check an integral, and vice versa.

Trap: Negative parts count positive: $(-2)^2 = 4$.

Trap: A squared ramp is a curve, not a triangle: $\frac{A^2D}{3}$, not $\frac{A^2D}{2}$.

Trap: If the graph repeats forever, $E_\infty = \infty$. Find the power over one period.



Worked: two flat pieces (left graph)

1. PIECES
flat at 2, $D=1$ flat at -1, $D=2$

↓

2. TEMPLATE
$E = \boxed{A^2D} + \boxed{A^2D}$

↓

3. PLACE
$E = 2^2(1) + (-1)^2(2) = 4+2 = 6$

Worked: a triangle (right graph)

1. PIECES
ramp $0 \to 2$, $D=1$ ramp $2 \to 0$, $D=1$

↓

2. TEMPLATE
$E = \frac{\boxed{A}^2\boxed{D}}{3} + \frac{\boxed{A}^2\boxed{D}}{3}$

↓

3. PLACE
$E = \frac{2^2(1)}{3} + \frac{2^2(1)}{3} = \frac{8}{3}$

↓

4. CHECK BY INTEGRAL
rise $x = 2t$: $\int_0^1 4t^2\,dt = \frac{4}{3}$; the fall is its mirror, $\frac{4}{3}$ ✓

Worked: a line from 1 to 3 over $[0,2]$

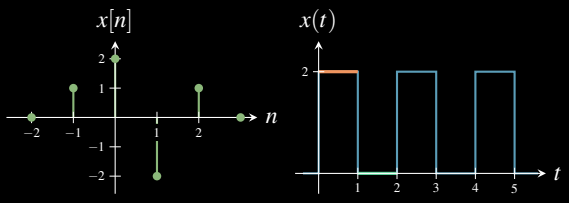
1. PIECES
$A=1$ $B=3$ $D=2$

↓

2. PLACE IN $\frac{\boxed{D}}{3}(\boxed{A}^2+\boxed{A}\boxed{B}+\boxed{B}^2)$
$\frac{2}{3}(1+1\cdot3+9) = \frac{26}{3}$

↓

3. CHECK BY INTEGRAL
$x = 1+t$: $\int_0^2 (1+2t+t^2)\,dt = 2+4+\frac{8}{3} = \frac{26}{3}$ ✓



Worked: a DT stem plot (left graph)

1. PIECES (EVERY STEM)
heights $1, 2, -2, 1$ at $n = -1, 0, 1, 2$

↓

2. PLACE IN $\Sigma\boxed{h}^2$
$E = 1+4+4+1 = 10$ *(no integral in DT)* $P = \lim_{N\to\infty} \frac{10}{2N+1} = 0 \Rightarrow$ energy signal

Worked: a periodic graph (right graph): power, not energy

1. PIECES (ONE PERIOD, $T=2$)
flat at 2, $D=1$ flat at 0, $D=1$

↓

2. TEMPLATE
$P = \frac{\boxed{A^2D} + \boxed{A^2D}}{T}$

↓

3. PLACE
$P = \frac{4+0}{2} = 2$ repeats forever: $E=\infty \Rightarrow$ power signal

6. transformations of the time variable

THE IDEA
You are not changing the shape, only *when* things happen. Ask one question of every corner of the graph: **where did this point move to?** It keeps its height and just lands at a new time.

- **Shift** $x(t-t_0)$: $t_0 > 0$ moves right (a **delay**); $t_0 < 0$ moves left (an **advance**).
- **Reversal** $x(-t)$: play the tape backwards, mirror about $t=0$.
- **Scaling** $x(at)$: fast-forward (compress) if $|a|>1$; slow motion (stretch) if $|a|<1$.

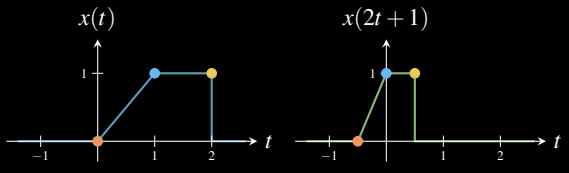
Point map for $x(at+b)$ — the template, one step, no order of operations:

$$t_{\text{new}} = \frac{\boxed{t_{\text{old}}} - b}{a}$$

Trap: Scale-then-shift is wrong: $x(2t)$ shifted left 1 is $x(2(t+1)) = x(2t+2)$, not $x(2t+1)$.

Trap: The sign inside is opposite to the direction: $x(t-3)$ moves *right*.

Trap: DT scaling loses photos: $x[2n]$ keeps only the even-index samples.



Worked: sketch $x(2t+1)$ from $x(t)$ (the dots move)

1. PIECES
breakpoints of x: $t=0$ (height 0), $t=1$ (height 1), $t=2$ (height 1, then drops)
$a=2$, $b=1$

↓

2. TEMPLATE
$t_{\text{new}} = \frac{\boxed{t_{\text{old}}} - 1}{2}$

↓

3. PLACE EACH DOT
$\frac{0-1}{2} = -\frac{1}{2}$ $\frac{1-1}{2} = 0$ $\frac{2-1}{2} = \frac{1}{2}$ *(heights unchanged)*

↓

4. CHECK
$t=0$: $x(2\cdot0+1) = x(1) = 1$ ✓ (the blue dot)

Worked: (reversal) sketch $x(-t+2)$ from the same $x(t)$

1. TEMPLATE
$a=-1$, $b=2$: $t_{\text{new}} = \dfrac{t_{\text{old}}-2}{-1} = 2-t_{\text{old}}$

↓

2. PLACE EACH DOT
$0\to 2$ $\quad 1\to 1$ $\quad 2\to 0$ $\Rightarrow$ flat 1 on $[0,1]$, ramp $1\to 0$ on $[1,2]$ (mirrored)

↓

3. CHECK
$t=1.5$: $x(0.5)=0.5$, on the ramp ✓

Worked: DT: $x[0]=1$, $x[1]=2$, $x[2]=3$, else 0

1. PLACE EACH SAMPLE WITH $n_{\text{new}} = \frac{n_{\text{old}}-b}{a}$
delay $x[n-2]$: $0,1,2\to 2,3,4$ $\quad$ reverse $x[-n]$: $0,1,2\to 0,-1,-2$
compress $x[2n]$: $0\to 0$, $\quad 2\to 1$, $\quad 1\to \frac{1}{2}$: not an integer, lost

Which way? — undo a transformation

Given → find	Do this
$x(t)\to x(at+b)$	map each breakpoint: $t_{\text{new}} = \frac{t_{\text{old}}-b}{a}$
$y(t)=x(at+b)\to x(t)$	$x(t)=y\left(\frac{t-b}{a}\right)$

Trap: $x[2n]$ cannot be undone. The odd-index photos were thrown away.

Worked: (reverse) $y(t)=x(2t+1)$; find $x(t)$

1. PIECES, TEMPLATE
$a=2$, $b=1$ $\quad x(t)=y\left(\dfrac{t-\boxed{b}}{\boxed{a}}\right)$

↓

2. PLACE, CHECK
$x(t)=y\left(\frac{t-1}{2}\right)$ $\quad x\left(2\cdot\frac{t-1}{2}+1\right)=x(t)$ ✓

Which way? — find $y=Ax(at+b)$ from two graphs

$$y(t)=\boxed{A}x\left(\boxed{a}t+\boxed{b}\right)$$

1. $A=\frac{\text{height of } y}{\text{height of } x}$; upside down means $A<0$.
2. Mirrored? Then $a<0$.
3. Two matching points: x at t_1, t_2 moved to τ_1, τ_2. Solve $a\tau_1+b=t_1$, $a\tau_2+b=t_2$ *(y at τ is x at aτ+b)*.
4. Check a third point.

Worked: x: ramp $0\to 1$ on $[0,4]$. y: same ramp, height 3, on $[-1,1]$

1. PIECES
$A=\frac{3}{1}=3$ $\quad$ not mirrored $\quad$ start $0\to -1$, $\quad$ end $4\to 1$

↓

2. SOLVE FOR a,b
$-a+b=0$, $\quad a+b=4$ $\Rightarrow$ add: $b=2$, $a=2$

↓

3. PLACE, CHECK
$y(t)=3x(2t+2)$ $\quad t=0$: $3x(2)=1.5$, half of 3 ✓

Worked: (mirrored) same x; y falls $3\to 0$ on $[-1,1]$

1. PIECES
$A=3$ $\quad$ mirrored $\quad$ start $0\to 1$, $\quad$ end $4\to -1$

↓

2. SOLVE, PLACE
$a+b=0$, $\quad -a+b=4$ $\Rightarrow a=-2$, $b=2$ $\quad y=3x(-2t+2)$

Worked: (DT) $x[0..2]=1,2,3$; $y[-1]=3$, $y[0]=2$, $y[1]=1$

1. PIECES
mirrored $\Rightarrow y[n]=x[-n+\boxed{b}]$ $\quad$ one match: $y[-1]=3=x[2]$

↓

2. SOLVE, CHECK
$1+b=2\Rightarrow b=1$: $\quad y[n]=x[-n+1]$ $\quad n=0$: $x[1]=2$ ✓

7. periodic signals

THE IDEA
A periodic signal is a song on repeat. Slide it by one period and nothing changes. The **fundamental period** is the shortest loop.

$$x(t)=x(t+T) \qquad x[n]=x[n+N],\ N \text{ an integer}$$

- **CT sinusoid**: always periodic, $\quad T=\dfrac{2\pi}{\boxed{\omega}}$.
- **DT sinusoid**: periodic only if a whole number of photos holds a whole number of turns: $\quad \dfrac{\boxed{\omega}}{2\pi}=\dfrac{m}{\boxed{N}}$ in lowest terms.
- **Sum of two**: the loops realign after $\text{lcm}(T_1,T_2)$.

Trap: $\cos(n)$ is not periodic ($\frac{1}{2\pi}$ is irrational), though $\cos(t)$ is. Sampling can destroy periodicity.

Worked: period of $\cos\left(\frac{2\pi}{3}t\right)+\sin\left(\frac{\pi}{2}t\right)$

1. PIECES
$\omega_1=\frac{2\pi}{3}$ $\quad \omega_2=\frac{\pi}{2}$

↓

2. PLACE IN $\frac{2\pi}{\boxed{\omega}}$
$T_1=\dfrac{2\pi}{2\pi/3}=3$ $\quad T_2=\dfrac{2\pi}{\pi/2}=4$

↓

3. LCM, CHECK
$\text{lcm}(3,4)=12$ s $\quad$ holds 4 and 3 whole cycles ✓

Worked: DT period: three cases

1. TEMPLATE
$\dfrac{\boxed{\omega}}{2\pi}=\dfrac{m}{\boxed{N}}$, reduced

↓

2. PLACE
$\cos\left(\frac{3\pi}{8}n\right)$: $\frac{3}{16}\Rightarrow N=16$ *($\frac{3\pi}{8}\cdot 16=3$ turns ✓)*
$\cos\left(\frac{4\pi}{6}n\right)$: $\frac{4}{12}=\frac{1}{3}\Rightarrow N=3$ not 12: reduce first
$\cos(1\,n)$: $\frac{1}{2\pi}$, not a fraction of integers $\Rightarrow$ not periodic

8. even and odd signals

THE IDEA
Even: a mirror on the vertical axis; the left half reflects the right. **Odd**: spin the graph half a turn about the origin and it lands on itself.

- Even: $x(-t)=x(t)$. $\quad \cos,\ t^2$, constants.
- Odd: $x(-t)=-x(t)$. $\quad \sin,\ t,\ t^3$.

Test: replace t with $-t$, simplify, compare. $\quad$ **Split**:

$$\mathcal{E}v\{x\}=\frac{x(t)+x(-t)}{2} \qquad \mathcal{O}d\{x\}=\frac{x(t)-x(-t)}{2}$$

Shortcut for sums of terms: sort each term into a even or odd pile; the piles *are* the two parts. $\quad$ Products work like signs: even·even = odd·odd = even; even·odd = odd.

Trap: An odd signal is 0 at the origin: $x(0)=-x(0)$.

Worked: is $x(t)=t^3-2t$ even, odd, or neither?

1. PIECES (SORT TERMS)
t^3 odd $\quad -2t$ odd $\quad$ no even terms

↓

2. CHECK BY TEST
$x(-t)=-t^3+2t=-(t^3-2t)=-x(t)\Rightarrow$ odd

Worked:* $x(t) = t^2 + t$*: test, then split

1. PIECES (SORT TERMS)
t^2 even t odd both piles non-empty $\Rightarrow$ neither

↓

2. PLACE
$\mathscr{E}v = t^2$ $\mathscr{O}d = t$

↓

3. CHECK BY FORMULA
$\frac{(t^2+t)+(t^2-t)}{2} = t^2$ ✓ $\frac{(t^2+t)-(t^2-t)}{2} = t$ ✓

***Worked: DT split:* $x[0]=1$, $x[1]=2$, $x[2]=3$, *else* 0**

1. TABLE: FILL $x[-n]$ BY READING x AT THE OPPOSITE INDEX

n	-2	-1	0	1	2
$x[n]$	0	0	1	2	3
$x[-n]$	3	2	1	0	0
$\frac{x[n]+x[-n]}{2}$	1.5	1	1	1	1.5
$\frac{x[n]-x[-n]}{2}$	−1.5	−1	0	1	1.5

↓

2. CHECK EACH COLUMN
even + odd $= x[n]$: $n=2$: $1.5+1.5=3$ ✓ $n=-2$: 0 ✓

9. complex exponentials

THE IDEA
$e^{j\theta}$ is an arrow of length 1 pointing at angle θ. Let $\theta = \omega t$ and it spins; its shadow on the horizontal axis is a cosine. Multiply by $e^{\sigma t}$ and it also grows or shrinks: a spiral.

Euler: $e^{j\theta} = \cos\theta + j\sin\theta$. $\cos\theta = \frac{e^{j\theta}+e^{-j\theta}}{2}$, $\sin\theta = \frac{e^{j\theta}-e^{-j\theta}}{2j}$
(two arrows spinning opposite ways).

CT form: $x(t) = Ce^{at}$, $C = \boxed{|C|}e^{j\boxed{\phi}}$ (start length, angle), $a = \boxed{\sigma} + j\boxed{\omega}$ (growth, spin). The template:

$$\mathrm{Re}\{Ce^{at}\} = \boxed{|C|}e^{\boxed{\sigma}t}\cos\left(\boxed{\omega}t + \boxed{\phi}\right)$$

Case	Real part looks like
$\omega = 0$	no spin: **real exponential**; grows if $\sigma > 0$, decays if $\sigma < 0$
$\sigma = 0$	pure spin: **sinusoid**, $T = \frac{2\pi}{\lvert\omega\rvert}$
both	spiral: **damped sinusoid**; envelope $\lvert C\rvert e^{\sigma t}$

DT form: $x[n] = C\alpha^n$; each step multiplies by α. $|\alpha| < 1$ decays, $|\alpha| > 1$ grows, $\alpha < 0$ flips sign each step.

DT frequencies repeat every 2π: $e^{j(\hat{\omega}+2\pi)n} = e^{j\hat{\omega}n}$, since $e^{j2\pi n} = 1$ for integer n.

CT $e^{j\omega t}$	DT $e^{j\hat{\omega}n}$
each ω a different signal	$\hat{\omega}$ and $\hat{\omega}+2\pi$ the same
always periodic	periodic only if $\frac{\hat{\omega}}{2\pi}$ is rational
larger $\omega \Rightarrow$ faster	fastest at $\hat{\omega} = \pi$ $(+1,-1,\ldots)$

***Worked: real part of* $x(t) = 2e^{j\pi/3}e^{(-1+j4)t}$**

1. PIECES
C: $|C|=2$, $\phi = \frac{\pi}{3}$ a: $\sigma=-1$, $\omega=4$

↓

2. PLACE
$\mathrm{Re}\{x\} = 2e^{-1t}\cos\left(4t+\frac{\pi}{3}\right)$ *(damped sinusoid, envelope $2e^{-t}$)*

***Worked: (reverse) write* $5e^{-2t}\cos\left(3t-\frac{\pi}{4}\right)$ *as* $\mathrm{Re}\{Ce^{at}\}$**

1. PIECES (MATCH THE TEMPLATE)
$|C| = 5$ $\sigma = -2$ $\omega = 3$ $\phi = -\frac{\pi}{4}$

↓

2. PLACE
$C = 5e^{j(-\pi/4)} = 5e^{-j\pi/4}$ $a = -2 + j3 = -2 + j3$

Worked: cosine to exponentials, and a DT alias

1. $6\cos(5t)$
$6 \cdot \frac{e^{j5t}+e^{-j5t}}{2} = 3e^{j5t} + 3e^{-j5t}$

↓

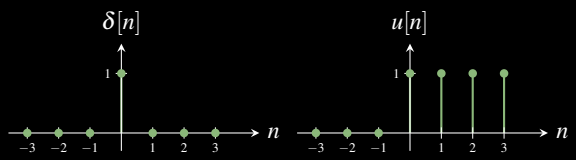
2. $\cos\left(\frac{9\pi}{4}n\right)$
remove a whole turn: $\frac{9\pi}{4} - 2\pi = \frac{\pi}{4} \Rightarrow \cos\left(\frac{\pi}{4}n\right)$, $\frac{\pi/4}{2\pi} = \frac{1}{8} \Rightarrow N = 8$

10. the button, the switch, and the ramp

THE IDEA
- δ is a **button**: press once at time 0, then release.
- u is a **switch**: flip it on at time 0 and leave it on.
- r is a **ramp**: the running total of the switch, climbing $0, 1, 2, 3, \ldots$

Each is the running total of the one before. Everything else is built from these three.

$$\delta[n] = \begin{cases} 1 & n = 0 \\ 0 & n \neq 0 \end{cases} \qquad u[n] = \begin{cases} 1 & n \geq 0 \\ 0 & n < 0 \end{cases}$$



Trap: The switch does not keep climbing: $u[n]$ is $0, 0, 1, 1, 1, \ldots$ One stair up, then flat. The ramp is what climbs.
Size: the button is one stem of height 1, $E_\infty = 1$ (energy signal). The switch stays on forever: $E_\infty = \infty$, finite power (power signal).

Running totals and where the button sits

Switch = running total of presses: $u[n] = \sum_{k=-\infty}^{n} \delta[k]$ *(no presses before 0; one press at 0; the total stays 1)*. **Back**: a switch on at 0 minus a switch on at 1 is on only at 0: $\delta[n] = u[n] - u[n-1]$.
Where is it? Set the inside to 0 and solve. $\delta[n-k]$ sits at $n=k$; $u[n-k]$ turns on at $n=k$.

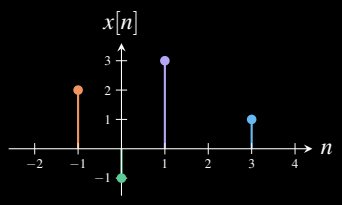
Inside	Stem at	Moves	Name
$n-3$	$n=3$	right	**delay**: later
$n+2$	$n=-2$	left	**advance**: earlier

Trap: The sign inside is opposite to the position, and $+2$ is an *advance*.

Building signals: switches on and off, one button per stem

Pulse: $u[n-\boxed{a}] - u[n-\boxed{b}]$ is on for $\boxed{a} \leq n \leq \boxed{b}-1$ ($b-a$ samples). CT: $u(t-t_1) - u(t-t_2)$ is on for $t_1 \leq t < t_2$.
Trap: The off-switch's own position is already off: $u[n] - u[n-4]$ at $n=4$ is $1-1=0$.
One button per stem: a stem of height h at position k is $\boxed{h}\,\delta[n-\boxed{k}]$. Add one term per stem:

$$x[n] = \sum_k x[k]\,\delta[n-k]$$



Worked: write the stems above with buttons

1. PIECES (HEIGHT AT POSITION)
2 at -1 $\quad -1$ at 0 $\quad$ 3 at 1 $\quad$ 1 at 3 *(no stem at 2: no term)*

↓

2. TEMPLATE (ONE PER STEM)
$\boxed{h}\,\delta[n-\boxed{k}]+\cdots$

↓

3. PLACE
$x[n]=2\delta[n+1]-\delta[n]+3\delta[n-1]+\delta[n-3]$

Worked: (reverse) read the stems of $x[n]=4\delta[n-2]-2\delta[n+3]$

1. PIECES
coefficient = height; $\quad$ inside $=0 \Rightarrow$ position

↓

2. PLACE, CHECK
$n-2=0$: $x[2]=4$ $\quad n+3=0$: $x[-3]=-2$ $\quad n=2$: $4(1)-2(0)=4$ ✓

Worked: DT pulse $x[n]=u[n]-u[n-3]$

1. PIECES
on at $a=0$, off at $b=3$

↓

2. PLACE IN $\boxed{a}\le n\le\boxed{b}-1$
on for $0\le n\le 2 \Rightarrow \delta[n]+\delta[n-1]+\delta[n-2]$

↓

3. FIRST DIFFERENCE
a button at each switch: $x[n]-x[n-1]=\delta[n]-\delta[n-3]$

The CT button

A one-instant press does not exist in CT. Picture a rectangle of width ε, height $\frac{1}{\varepsilon}$: area 1 always. Shrink $\varepsilon\to 0$: infinitely tall, area still 1. A hammer tap: instant, but a definite amount of hit.

$$\int_{-\infty}^{\infty}\delta(t)\,dt=1 \qquad \text{(CT: area 1; DT: height 1)}$$

CT switch: $u(t)=\int_{-\infty}^{t}\delta(\tau)\,d\tau$; back: $\delta(t)=\frac{du}{dt}$. A jump in a signal becomes a button in its derivative.
Trap: δ is a function of t, not a variable. $\int\delta\,dt$ is not δ^2; use its one fact, area 1.

Sifting: the camera flash

Fire a flash at t_0 and see what value of g it captures. The template:

$$\int_{\boxed{\text{lo}}}^{\boxed{\text{hi}}}\boxed{g(t)}\,\delta\big(t-\boxed{t_0}\big)\,dt=\begin{cases}\boxed{g}(\boxed{t_0}) & \boxed{t_0}\text{ inside }[\boxed{\text{lo}},\boxed{\text{hi}}]\\ 0 & \text{outside}\end{cases}$$

Three questions: (1) Where is the flash? Set the inside of δ to 0. (2) In the window? If not, the answer is 0. (3) What does it capture? Plug t_0 into everything except the δ. The answer has no δ left.
Sampling form: $x(t)\,\delta(t-t_0)=x(t_0)\,\delta(t-t_0)$; DT the same with n_0.

Worked: five flashes

1. PLACE: FLASH / WINDOW / CAPTURE
$\int_{-2}^{6}(3t+1)\delta(t-4)\,dt$: $\quad 4$ / in $[-2,6]$ / $3(4)+1=13$
$\int_{1}^{3}5\,\delta(t)\,dt$: $\quad 0$ / outside $[1,3]\Rightarrow 0$ *(over all t it would be 5)*
$\int_{-\infty}^{\infty}(t^2+1)\delta(t-2)\,dt$: $\quad 2$ / in / $2^2+1=5$
$\int_{0}^{5}e^t\,\delta(t+1)\,dt$: $\quad -1$ / outside $[0,5]\Rightarrow 0$
DT $\sum_n n^2\delta[n-3]$: $\quad 3$ / in / $3^2=9$

Which way? — switch or button

Given → find	Do this
$u\to\delta$ (CT)	differentiate: $\delta(t)=\dfrac{du}{dt}$
$\delta\to u$ (CT)	integrate: $u(t)=\int_{-\infty}^{t}\delta(\tau)\,d\tau$
$u\to\delta$ (DT)	first difference: $\delta[n]=u[n]-u[n-1]$
$\delta\to u$ (DT)	running sum: $u[n]=\sum_{k=-\infty}^{n}\delta[k]$

Trap: CT uses derivatives and integrals. DT uses differences and sums.

Worked: pulse of height 2 on $[1,4)$ with switches, then down and back

1. PIECES
jump up 2 at $t=1$ $\quad$ jump down 2 at $t=4$

↓

2. TEMPLATE (ONE SWITCH PER JUMP)
$\boxed{\text{jump}}\cdot u(t-\boxed{\text{where}})+\cdots$

↓

3. PLACE, CHECK
$x(t)=2u(t-1)-2u(t-4)$ $\quad t=2$: 2 ✓ $\quad t=5$: 0 ✓

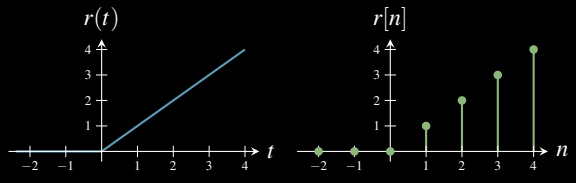
↓

4. DOWN AND BACK
$\dfrac{dx}{dt}=2\delta(t-1)-2\delta(t-4)$ *(each switch becomes a button)*; integrate to return ✓

The ramp: the running total of the switch

$$r(t)=t\,u(t)=\begin{cases}t & t\ge 0\\ 0 & t<0\end{cases} \qquad r[n]=n\,u[n]$$

It grows without limit: neither energy nor power.



The chain: button δ $\xrightarrow{\text{running total}}$ switch u $\xrightarrow{\text{running total}}$ ramp r. Back: differentiate (CT) or first difference (DT).

Form	Formula
CT: integral of the switch	$r(t)=\int_{-\infty}^{t}u(\tau)\,d\tau$
CT: derivative of the ramp	$\dfrac{dr}{dt}=u(t)$
DT: running sum of the switch	$r[n]=\sum_{k=-\infty}^{n-1}u[k]$
DT: first difference	$r[n]-r[n-1]=u[n-1]$
DT: sum of shifted switches	$r[n]=\sum_{k=1}^{\infty}u[n-k]$
DT: sum of buttons	$r[n]=\sum_{k=0}^{\infty}k\,\delta[n-k]$

Trap: The DT sum stops at $n-1$: $r[1]=1$ adds only $u[0]$. Stopping at n gives $r[n+1]$.

Worked: check the DT forms at a few n

1. RUNNING SUM $\sum_{k\le n-1}u[k]$
$n=0$: nothing $=0$ ✓ $\quad n=1$: $u[0]=1$ ✓ $\quad n=3$: $u[0]+u[1]+u[2]=3$ ✓

↓

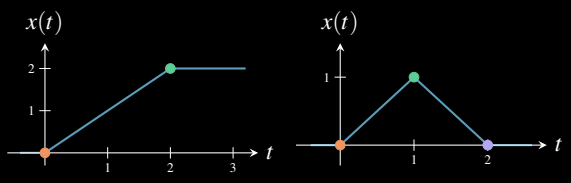
2. FIRST DIFFERENCE
$r[0]-r[-1]=0,\quad r[1]-r[0]=1,\quad r[2]-r[1]=1\Rightarrow u[n-1]$

Graphs from ramps and switches

Walk left to right. At each corner, add a ramp weighted by the *change in slope*; at each jump, a switch weighted by the jump:

$$x(t)=\sum\boxed{\Delta\text{slope}}\,r\big(t-\boxed{\text{corner}}\big)+\sum\boxed{\text{jump}}\,u\big(t-\boxed{\text{where}}\big)$$

$A\,r(t-t_0)$ is 0 before t_0, then climbs with slope A.



Worked: rise $0 \to 2$ on $[0,2]$, then flat (left graph)

1. PIECES (SLOPE BEFORE → AFTER)
$t=0$: $0 \to 1$, change $+1$ $\quad$ $t=2$: $1 \to 0$, change -1

↓

2. PLACE, CHECK
$x(t) = r(t) - r(t-2)$ $\quad$ $t=1$: 1 ✓ $\quad$ $t=5$: $5-3=2$ ✓

Worked: a triangle (right graph)

1. PIECES
$t=0$: $0 \to 1$, $+1$ $\quad$ $t=1$: $1 \to -1$, -2 $\quad$ $t=2$: $-1 \to 0$, $+1$

↓

2. PLACE, CHECK
$x(t) = r(t) - 2r(t-1) + r(t-2)$ $\quad$ $t=3$: $3-4+1=0$ ✓ *(slope changes sum to 0)*

Worked: (reverse) sketch $2r(t-1) - 2r(t-3)$; then a jump plus ramps

1. READ THE PIECES
corner at 1: slope $+2$ $\quad$ corner at 3: slope back to $0 \Rightarrow 0$, then rises to $2(2) = 4$, then flat at 4

↓

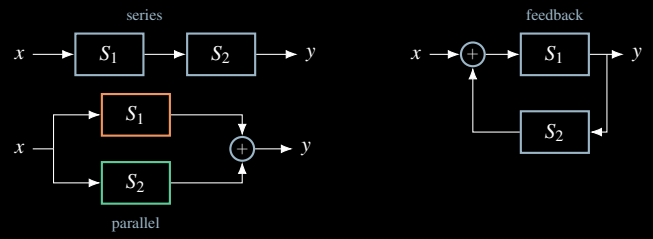
2. JUMP OF 3 AT 0, THEN SLOPE -1 UNTIL 0
jump $+3$ at 0, $\quad$ corner at 0: -1, $\quad$ corner at 3: $+1$ *(3 − t = 0)*
$x(t) = 3u(t) - r(t) + r(t-3)$

11. systems and interconnections

THE IDEA
A **system** is a machine: a signal goes in, a different one comes out, $x \to y$. A **difference equation** is the machine's recipe: how to cook $y[n]$ from inputs, and sometimes from its own past outputs.

Three hookups:

- **Series**: an assembly line. S_1's output is S_2's input: substitute S_1 into S_2's input slot.
- **Parallel**: x feeds both; outputs add: $y = \boxed{y_1} + \boxed{y_2}$.
- **Feedback**: the output loops back through S_2 into the input. The machine hears its own past.



Worked: series: S_1: $w[n] = 2x[n]$, then S_2: $y[n] = w[n] - w[n-1]$

1. PIECES
$w[n] = 2x[n]$, so $w[n-1] = 2x[n-1]$ *(replace every n with n − 1)*

↓

2. PLACE IN $y[n] = \boxed{w[n]} - \boxed{w[n-1]}$
$y[n] = 2x[n] - 2x[n-1]$

Worked: parallel: S_1: $2x[n]$, $\quad S_2$: $x[n-1]$

1. PLACE IN $\boxed{y_1} + \boxed{y_2}$
$y[n] = 2x[n] + x[n-1]$

Worked: feedback: $y[n] = x[n] + 0.5\,y[n-1]$; press the button $x = \delta$, $y[-1] = 0$

1. RUN IT, ONE STEP AT A TIME
$y[0] = 1 + 0.5(0) = 1$ *(the press)* $\quad$ $y[1] = 0 + 0.5(1) = 0.5$ *(only the echo)* $\quad$ $y[2] = 0.25$
$y[3] = 0.125$

↓

2. PATTERN
$y[n] = (0.5)^n\, u[n]$ *(each echo is half the last)*

DT systems as difference equations

System	Difference equation	What it does
unit delay	$y[n] = x[n-1]$	one step late
gain	$y[n] = a\,x[n]$	volume knob
first difference	$y[n] = x[n] - x[n-1]$	change since last
3-point moving average	$y[n] = \frac{1}{3}\sum_{k=0}^{2} x[n-k]$	smooths
accumulator	$y[n] = \sum_{k=-\infty}^{n} x[k]$	running total
accumulator, recursive	$y[n] = y[n-1] + x[n]$	same total
bank balance	$y[n] = 1.01\,y[n-1] + x[n]$	1% interest + deposit

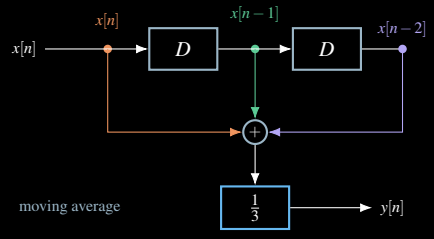
The accumulator is the running-total machine: button in, switch out; switch in, ramp out.

Nonrecursive (FIR): inputs only. **Recursive** (IIR): also its own past outputs; needs a start value like $y[-1] = 0$.

Block diagram parts: adder (circle) = sum of inputs; $\quad$ gain (box a) = a·input; $\quad$ unit delay (box D) = one-step memory, $x[n] \to x[n-1]$; $\quad$ branch dot copies a signal.

Diagram ↔ equation: each wire is a piece. One D per shift, one gain per coefficient, one adder per sum.

Trap: A D that feeds $y[n]$ back into the adder means feedback: the equation has $y[n-1]$ (recursive).



Worked: (diagram to equation) read the moving-average diagram

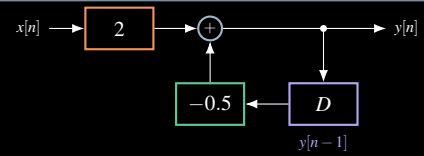
1. PIECES (NAME EACH WIRE INTO THE ADDER)
$x[n]$ $\quad$ after one D: $x[n-1]$ $\quad$ after two: $x[n-2]$ $\quad$ gain $\frac{1}{3}$

↓

2. TEMPLATE
$y[n] = \boxed{\text{gain}}\,(\boxed{\text{wire}} + \boxed{\text{wire}} + \boxed{\text{wire}})$

↓

3. PLACE
$y[n] = \frac{1}{3}\,(x[n] + x[n-1] + x[n-2])$



Worked: (equation to diagram) $y[n] = 2x[n] - 0.5\,y[n-1]$

1. PIECES (ONE BLOCK PER PIECE)
input gain 2 $\quad$ $y[n-1]$: one D on the output side (feedback) $\quad$ its gain -0.5

↓

2. PLACE (ABOVE)
$x \to [2] \to (+) \to y$; $\quad$ from y: dot $\to [D] \to [-0.5] \to$ back into $(+)$

Worked: run the moving average: $x[0..2] = 3, 6, 9$, others 0

1. SLIDE A 3-SAMPLE WINDOW, DIVIDE BY 3
$y[0] = \frac{3}{3} = 1 \quad y[1] = \frac{6+3}{3} = 3 \quad y[2] = \frac{9+6+3}{3} = 6 \quad y[3] = \frac{9+6}{3} = 5 \quad y[4] = 3$ then 0

Worked: run the accumulator $y[n] = y[n-1] + x[n]$ with the switch, $y[-1] = 0$

1. RUN IT
$y[0] = 0+1 = 1 \quad y[1] = 1+1 = 2 \quad y[2] = 3 \Rightarrow (n+1)u[n] = r[n+1]$ *(running total of a switch: a ramp)*

↓

2. CHECK WITH THE SUM FORM
$y[2] = u[0] + u[1] + u[2] = 3$ ✓

12. system properties

THE IDEA
Properties are yes/no questions about a machine. Each is separate: a machine can pass one and fail another.

Property	In plain English
memoryless	It only looks at *now*.
causal	It cannot see the future; it may remember the past.
stable (BIBO)	It does not blow up: input under a ceiling ⇒ output under a ceiling.
invertible	You can undo it: different inputs give different outputs.
time invariant	It ignores the clock: delay in ⇒ same output, delayed.
linear	Scale in → scale out, add in → add out: $ax_1 + bx_2 \to ay_1 + by_2$.

Quick red flags

Trap: Linearity: put in $x = 0$; a linear machine gives $y = 0$. The $+3$ in $2x+3$ fails.
Trap: Time invariance: an n or t *outside* $x(\cdot)$ ($nx[n]$) reads the clock. Scaling or flipping inside ($x(2t)$, $x(-t)$) also fails.
Trap: Causality: an advance like $x[n+1]$. And $y(t) = x(-t)$: at $t = -1$ it needs $x(1)$.
Trap: Stability: an n multiplying x, or a sum with no end, can blow up.

Memory and causality: read the time stamps

Every $x[\cdot]$ on the right carries a time stamp. Color each one: now (n), past ($n-1, \ldots$), future ($n+1, \ldots$).

memoryless $\iff$ every stamp is now causal $\iff$ no stamp is future

Worked: memory, causality, invertibility

1. PLACE THE STAMPS
$y[n] = x[n] + x[n-1]$: a past stamp ⇒ memory, causal
$y[n] = x[n+1]$: a future stamp ⇒ not causal

↓

2. INVERTIBLE? TRY TO SOLVE FOR x
$y = 2x$: $x = \frac{y}{2}$ invertible $y = x^2$: $\pm 1 \to 1$ not invertible *(the sign is lost)*

Time invariance: two paths, then compare

Path 1: delay the input, then run the machine — change only the inside of x.
Path 2: run the machine, then delay the output — change *every* n, including any lone n. Same result ⇒ time invariant.

Worked: three machines

1. PLACE: PATH 1 vs PATH 2
$nx[n]$: $nx[n-n_0]$ vs $(n-n_0)x[n-n_0]$ ⇒ time varying *(the lone n)*
$x(2t)$: $x(2t-t_0)$ vs $x(2t-2t_0)$ ⇒ time varying
$x[n-1]$: $x[n-1-n_0]$ vs $x[n-n_0-1]$ ⇒ time invariant

Linearity: combine first, or combine after?

Path 1: mix the inputs ($ax_1 + bx_2$), then run. **Path 2**: run each, then mix the outputs. Linear means they agree.

Worked: $y = 2x+3$ and $y = nx[n]$

1. $2x+3$
quick test: $x = 0 \to 3 \neq 0$. Path 1: $2ax_1 + 2bx_2 + 3$ vs Path 2: $2ax_1 + 2bx_2 + 3a + 3b$ ⇒ not linear *(incrementally linear)*

↓

2. $nx[n]$
Path 1: $anx_1 + bnx_2$ = Path 2 ⇒ linear *(yet time varying: separate properties)*

Stability with a ceiling B

THE IDEA
BIBO: bounded in, bounded out. "Bounded" means a ceiling: $|x[n]| \le B < \infty$. Stable means the output has a ceiling too. This is Lesson 1 again: is the signal's size finite?

To show stable: take $|\cdot|$ of the equation, split with $|a+b| \le |a| + |b|$ and $|cx| = |c||x|$, then replace each $|x[\cdot]|$ (shifted ones too) with B:

$$|y| \le \boxed{|c_1|}B + \boxed{|c_2|}B + \cdots$$

A finite number (no n, no endless sum) means stable. **To show unstable**: one capped input (like $x = 1$ or $u[n]$) with an unbounded output.
Trap: $|nx[n]| \le |n|B$ grows with n. The accumulator adds another B every step.

Worked: five machines

1. PLACE THE COEFFICIENTS
$2x[n] + 1x[n-1]$: $\le 2B + 1B = 3B \Rightarrow$ stable
moving average: $\le \frac{1}{3}(B+B+B) = B \Rightarrow$ stable
$e^{x[n]}$: $-B \le x \le B \Rightarrow |y| \le e^B \Rightarrow$ stable

↓

2. COUNTEREXAMPLES
$nx[n]$ with $x = 1$: $y = n$, unbounded ⇒ not stable
accumulator with $u[n]$: $y = n+1 \Rightarrow$ not stable

↓

3. RECURSIVE $y = 0.5y[n-1] + x$
unroll: $y[n] = \sum_{k \ge 0}(0.5)^k x[n-k] \Rightarrow |y| \le B \cdot \frac{1}{1-0.5} = 2B$ stable *(with 1.01 the echoes grow: unstable)*

Properties of common DT systems

System	Mless	Causal	Linear	TI	Stable
$x[n-1]$	no	yes	yes	yes	yes
$x[n] - x[n-1]$	no	yes	yes	yes	yes
moving average	no	yes	yes	yes	yes
accumulator	no	yes	yes	yes	no
$nx[n]$	yes	yes	yes	no	no
$x[n]^2$	yes	yes	no	yes	yes
$2x[n]+3$	yes	yes	no	yes	yes
$x[n+1]$	no	no	yes	yes	yes
$x[2n]$	no	no	yes	no	yes
$x[-n]$	no	no	yes	no	yes
$0.5y[n-1] + x[n]$	no	yes	yes	yes	yes

Mless = memoryless; TI = time invariant; the recursive system starts at rest.
Trap: The properties are separate: linear but time varying ($nx[n]$); stable but not causal ($x[n+1]$).